

Q.6	Answer the following:	[12]
(a)	Consider a PDA $M = (\{q_0, q_1, q_2, q_f\}, \{a, b\}, \{Z, A, B\}, \delta, q_0, Z, \{q_f\})$, the mapping δ is defined as 1. $\delta(q_0, b, Z) = \{(q_f, Z)\}$, 2. $\delta(q_0, a, Z) = \{(q_0, AZ)\}$, 3. $\delta(q_0, a, A) = \{(q_0, AA)\}$, 4. $\delta(q_0, b, A) = \{(q_1, A)\}$, 5. $\delta(q_1, b, A) = \{(q_2, \epsilon)\}$, 6. $\delta(q_2, b, A) = \{(q_1, A)\}$, 7. $\delta(q_2, b, Z) = \{(q_f, Z)\}$. i. What is the language recognized by the given PDA? ii. Construct an equivalent PDA acceptance by empty store $N(M')$ which will recognize the same language. iii. Whether the given PDA is deterministic or nondeterministic PDA? Justify your answer.	CO1[H] CO2[H] CO3[H]
(b)	Consider a grammar $G = (\{S, A, B\}, \{a, b\}, P, S)$ with production rule $S \rightarrow aB, B \rightarrow b, B \rightarrow bS, B \rightarrow aBB, S \rightarrow bA, A \rightarrow a, A \rightarrow aS, A \rightarrow bAA$. With inductive hypothesis prove that if a string is derivable from S then it has equal number of a's and b's also prove that if string has equal number of a's and b's then it can be derivable from S.	CO1[H] CO2[H] CO3[H] CO4[M]
Q.7	Answer the following:	[12]
(a)	Give a context free grammar (CFG) G without ϵ - production to generate the language $L = \{a^n b^m c^n \mid m, n \geq 1\}$. Design a push down automata M corresponding to the CFG G and show that the string $aaab^1bccc$ is accepted by M.	CO1[H] CO2[H] CO5[H]
(b)	Given a PDA $M = (\{q_1, q_2\}, \{0, 1\}, \{Z, X\}, \delta, q_1, Z, \emptyset)$. The transition function δ is defined by: 1. $\delta(q_1, 0, Z) = \{(q_1, XZ)\}$, 2. $\delta(q_1, 0, X) = \{(q_1, XX)\}$, 3. $\delta(q_1, 1, X) = (q_2, \epsilon)$, 4. $\delta(q_2, 1, X) = \{(q_2, \epsilon)\}$, 5. $\delta(q_2, \epsilon, X) = (q_2, \epsilon)$, 6. $\delta(q_2, \epsilon, Z) = \{(q_2, \epsilon)\}$ What is $N(M)$? Convert the given PDA M to CFG G. Simplify the resultant grammar G by removing useless productions.	CO1[H] CO2[H] CO3[H] CO4[M]
Q.8	Answer the following:	[12]
(a)	What is the language generated by the following grammar $G = (\{S, A, B\}, \{a, b\}, \{S \rightarrow aA, S \rightarrow bA, A \rightarrow aB, A \rightarrow bB, B \rightarrow aS, B \rightarrow bS, B \rightarrow a, B \rightarrow b\}, S)$? Construct the equivalent deterministic finite state automata that recognize the language generated by the grammar G.	CO1[H] CO2[M] CO5[H]
(b)	Prove that if a context free language L is accepted by a push down automata M1 with empty store then the same language can be recognized by the push down automata M2 with final state.	CO1[H] CO2[H]
(c)	Explain Church Turing Hypothesis. Define recursive and recursively enumerable languages and specify relationship between them.	CO1[H]
Q.9	Answer the following:	[12]
(a)	Why Turing machine is more powerful than that of FA and PDA? With suitable example design and illustrates a Turing machine as i. An input output device. ii. A computational device. iii. A language recognizer/accepting device.	CO1[H] CO2[M] CO3[M] CO5[H]
(b)	Design a Turing machine for the function $f(x) = x^2$. Show an instantaneous description for $x = 3$.	CO1[H] CO5[H]
